



UNIVERSITEIT•STELLENBOSCH•UNIVERSITY
jou kennisvennoot • your knowledge partner

Procedure for Workshop Work in MMW

for

Mechanical Project 478

Mechatronic Project 478

Postgraduate Students

Workshop work for students doing Mechatronic Project 488 (i.e. projects lead by E&E) is arranged by E&E.

NB: Workshop work must be planned well in advance and discussed with laboratory technicians. Many persons call on the workshop's time and the workshop is particularly busy during the second semester. Further, it usually takes more than 1 week to obtain the material required for a task. Therefore, ensure that you included enough lead time in your planning for workshop work and get your work request to the workshop as early as possible. Jobs are done according the following priorities (decreasing order): practicums, undergraduate projects, postgraduate projects, research projects and then ITM projects, but jobs already in process will usually not be stopped for new jobs.

Requirements for Drawings

- Drawings must be on scale 1:1 as far as possible, but not smaller than 1:2. The department's title block must be on the drawing. Drawings may not be made on sheets smaller than A3.
- The responsible student's name, student number and a contact telephone number must be given on each drawing.
- The drawing pack MUST have a parts list with the assembly drawings attached. A complete materials list will be appreciated.

Hints for Designs

- Use processes such as laser cutting (see document "Procedures for Laser Cutting") to save cost and time. Off-the-shelf components are also usually cheaper than making a similar thing in the workshop.
- Students must ask workshop staff's advice on material selection and manufacturability rather sooner than later. The later changes are made, the more work has to be redone!

- Mistakes on drawings and incomplete information can cause considerable delays in the completion of your work request. Therefore make sure that your drawings are correct and complete.
- See the list of available material (<http://mecheng.sun.ac.za>) in the material store and, where applicable make use of this as much as possible in your design. This will save time and work on your project can start earlier.
- If you plan to make use of CNC machining, see the workshop technicians early in the process to discuss the design and procedures to follow.

Procedures for Work Request

1. Complete the manufacturing drawings (see requirements above).
2. Obtain the study leader's approval of the design (study leader signs the parts list).
3. Arrange a meeting with Mr Cobus Zietsman (cjz@sun.ac.za) or Mr Ferdi Zietsman (cfz@sun.ac.za) for checking the design and preparing a cost estimate. The workshop staff will decide what material must be purchased and what can be taken from the material store.
4. Receive the list of materials to be purchased from Mr Zietsman and follow the "Procedures for Purchasing from Outside Firms" (online document). Ensure that the material has been received by MMW after delivery. Keep Ms Merentia Meyer informed of when deliveries can be expected.
5. Your task will only be started if the design has been completed in full, all drawings handed in at the workshop and all the material has arrived or is available.
6. When your task has been started, visit MMW once per day in case the artisan has questions about your design or encountered problems. Your job will be much delayed if they must first look for you, since they will start another job. Your task will be reregistered; therefore you will have to fall in the queue again. Regular interaction with the workshop staff also gives you the ideal opportunity to improve your knowledge of workshop practice.
7. After the work has been completed, the actual cost for materials from the store will be transferred by Ms Meyer from the relevant cost points.
8. The final labour and material costs will be send to you via email or you can get this information from Me Maurisha Galant at the store in the workshop. This should be included in the budget as reported in your final report.

Small Work Requests

Tasks that require less than 1 hour workshop time and that do not require material purchases, can at Mr Zietsman's discretion, be done without first obtaining the study leader's approval.